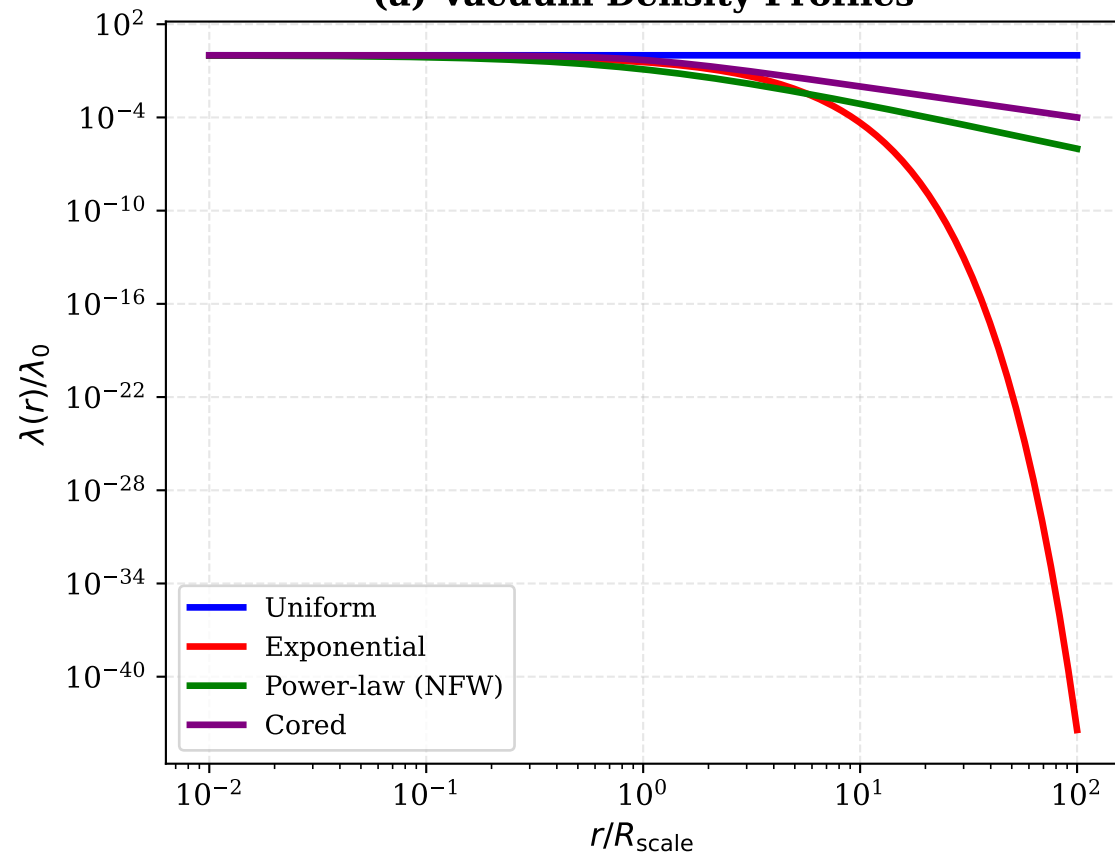
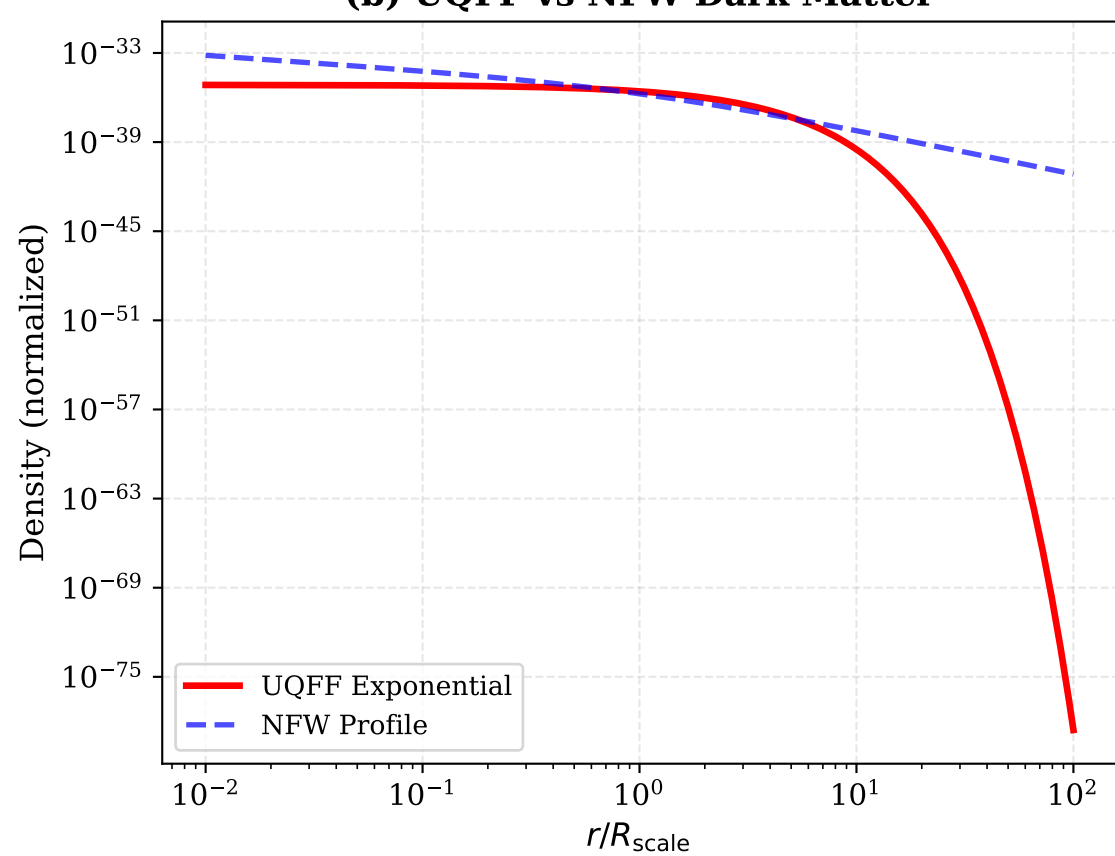
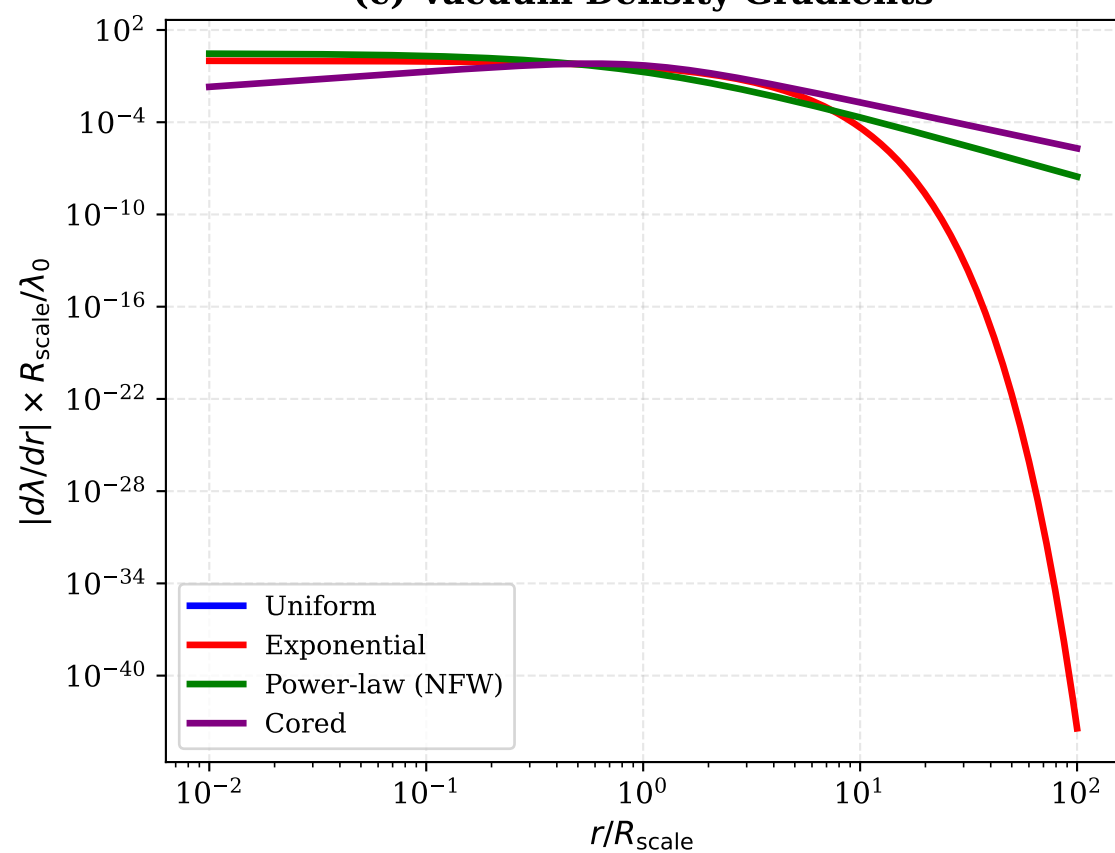


(a) Vacuum Density Profiles**(b) UQFF vs NFW Dark Matter****(c) Vacuum Density Gradients**

Physical Scales and Interpretation

Exponential Profile: $\lambda(r) = \lambda_0 e^{-r/R}$

Scale lengths:

- $R_{\text{scale}} = 10^{26} \text{ m} \approx 100 \text{ kpc}$ (galactic)
- $\lambda_0 = 7.12 \times 10^{-36} \text{ J/m}^3$ (UQFF vacuum)

Characteristic radii:

- $r = 0.1R$: $\lambda = 90\% \lambda_0$ (core)
- $r = 1.0R$: $\lambda = 37\% \lambda_0$ (half-light)
- $r = 10R$: $\lambda = 0.005\% \lambda_0$ (halo edge)

Comparison with NFW:

Both profiles exhibit $\sim r^{-3}$ power-law decay at large radii, suggesting vacuum energy clustering may contribute to dark matter rotation curves.

Observational Tests:

- SPARC galaxy rotation curves
- Weak lensing profiles
- X-ray cluster profiles